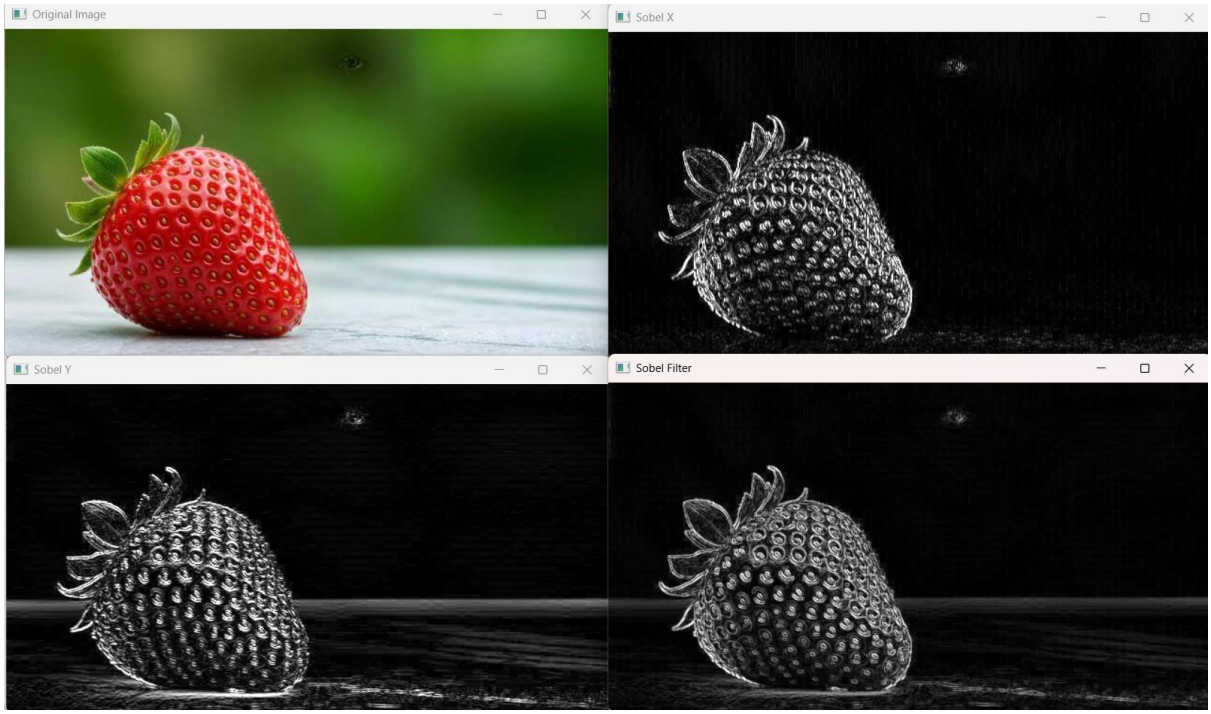
16.Implement a Sobel algorithm using Open CV to filter the input image.

CODE:

```
import cv2
img = cv2.imread(r"C:\Users\SUTHISH\Downloads\images.jpg")
if img is None:
   print("Image not found")
else:
   gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
   sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
   sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
   sobelx = cv2.convertScaleAbs(sobelx)
   sobely = cv2.convertScaleAbs(sobely)
   result = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)
   cv2.imshow("Original Image", img)
   cv2.imshow("Sobel X", sobelx)
   cv2.imshow("Sobel Y", sobely)
   cv2.imshow("Sobel Filter", result)
   cv2.waitKey(0)
   cv2.destroyAllWindows()
```

OUTPUT:



: